Atharva R Ankad

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Education

VIT Bhopal University — Bhopal, India
Integrated M.Tech Computer Science and Engineering (Computational and Data Science) – CGPA: 7.83 — *2023-2028*

Sri Chaitanya Techno School — Bangalore, India
CBSE Class XII Science – 75.6% — *2021-2023*

HAL Gnanajyoti School — Bangalore, India
ICSE Class X – 89.8% — *2009-2021*

Technical Skills

Languages: Python, Java, SQL
ML / AI: Machine Learning, NLP, Retrieval-Augmented Generation (RAG), Transformers, scikit-learn, SentenceTransformers, LangChain
Backend & APIs: FastAPI, REST API Design
Data & Libraries: Pandas, NumPy, MongoDB, Hadoop (Fundamentals)
Tools & Platforms: Git, Linux
Concepts: Data Engineering, Data Integration, Vector Search, Pipeline Design

Projects

Fasal Sahayak AI | *Python, FastAPI, scikit-learn, RandomForest, Pandas, NASA POWER API* — GitHub

- Built an end-to-end ML pipeline and REST API for **data-driven crop yield prediction and agronomic advisory, integrating 13 heterogeneous data sources** including 16K+ historical crop records, live meteorological data (NASA POWER API), and soil research from ICAR and government publications.
- Designed a **two-stage RandomForestRegressor pipeline** separating outlier crops (e.g., sugarcane) from annual crops, applying log1p target normalization to handle skewed yield distributions.
- Developed a **dual-mode advisory engine** providing generalized agronomic guidance for standard users and threshold-based warnings for farmers entering Soil Health Card (SHC) metrics.
- Reduced query latency by implementing **lightweight JSON lookup stores** for pre-computed geographic data, eliminating runtime database calls.

RAG Semantic Search System | *Python, LangChain, SentenceTransformers, MongoDB, Groq API* — GitHub

- Built a high-performance **Retrieval-Augmented Generation (RAG)** pipeline that ingests PDF documents, generates local semantic embeddings using **SentenceTransformers (all-MiniLM-L6-v2)**, and performs vector similarity search via **MongoDB Atlas Vector Search**.
- Implemented context-aware **smart chunking** with LangChain text splitters to preserve paragraph structure and semantic meaning across document chunks.
- Integrated **Groq API** with a strict contextual system prompt to ensure LLM responses are grounded solely in retrieved document context, with citation support including page numbers and relevance scores.

Achievements

Smart India Hackathon (SIH) 2025 — Internal Nominee — VIT Bhopal University
Selected among the top 45 out of 949 idea submissions in the university's internal hackathon — *2024*

Certifications

Google IT Support Certificate — Google
Foundations of IT support, networking, and systems administration — *2026*

Building RAG Apps using MongoDB — MongoDB
Credential in Retrieval-Augmented Generation application development — *2026*